With 750 labelled tweets, does fine-tuning beat a calibrated open LLM?

Research question (unchanged)

If we only have **750 labelled tweets**, does fine-tuning **BERTurk** give a higher **macro-F1** on the official OffensEval-TR 2020 test set (3,528 tweets) than prompting an open LLM (**Gemma-2-9B-it** or **Qwen3-8B**) that uses the same 750 tweets: 8 as prompt examples, and the other 742 to choose its decision threshold?

The answer is no if

against either LLM, BERTurk's average over 10 runs is not at least **0.02** higher than the LLM's average over 3 runs, or a paired bootstrap test says the gap could be chance.

0.763

TF-IDF baseline on the frozen test set (95% CI 0.745–0.779)

60

baseline errors audited, grouped into categories with counts

0.790

Gemma-4-12B few-shot dev macro-F1 after threshold calibration

Vesile Han · Gürkan Göztepeli (CSE 654, research leads) · Mert Uğur · Utku Gökçek (CSE 484). 10 talk slides + 4 Project Card appendix slides.

30,000 train / 1,756 dev / 3,528 frozen test; 19.3% offensive

File / split	Tweets	NOT	OFF	OFF %
Official training file	31,756	25,625	6,131	19.3
Our train (seed 42)	30,000	24,208	5,792	19.3
Our dev	1,756	1,417	339	19.3
Official test (frozen)	3,528	2,812	716	20.3

OffensEval-TR 2020 (Çöltekin, LREC 2020). Annotations CC BY 2.0. @USER/URL already masked; no extra cleaning. 9 duplicate train texts; 1 train/test overlap.

REAL EXAMPLE · OFF · TRAIN ID 33518

“Kırmızı ağır karar Allah belanı versin”

“A red card is a harsh call, God damn you.” Swearing at a decision, not a person; still OFF.

CONTRAST · NOT

“…annem alacağı arabadan kaç kere vazgeçti ya kahretsin zalım trafik”

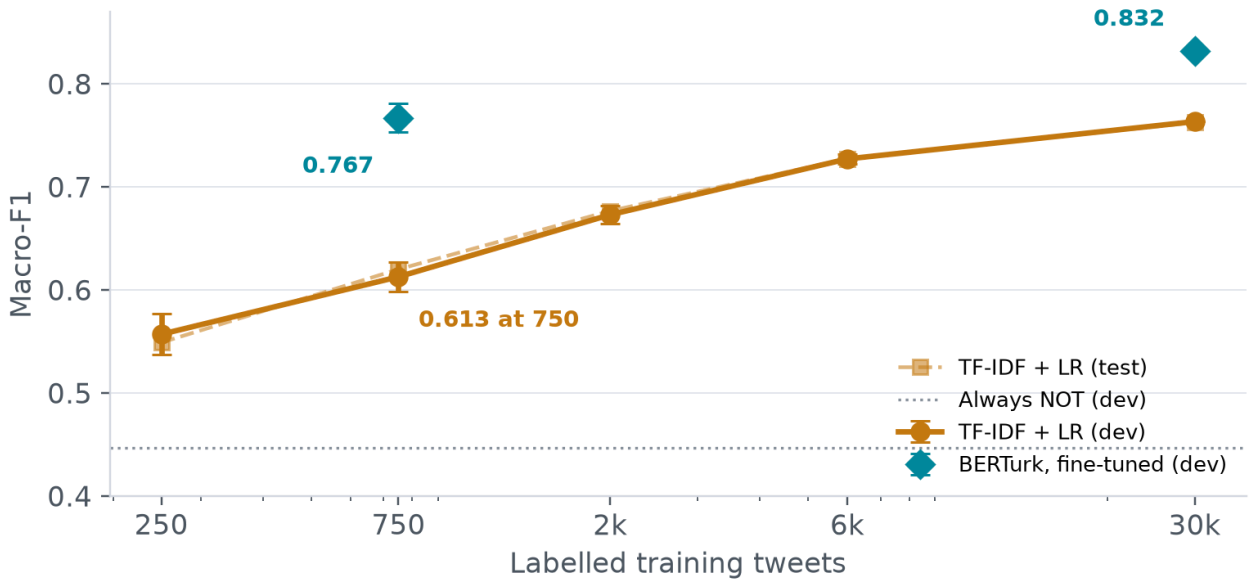
“How many times my mum gave up on buying a car, damn this cruel traffic.” Cursing at traffic is not offensive; target matters.

Why these counts matter. Only 19.3% of training tweets are offensive. Always saying “not offensive” is 80.7% accurate but scores 0.447 macro-F1, so accuracy hides the failure; we use macro-F1. Test frozen; dev only for code checks and mid-project numbers; every system uses the saved subsets (seed 42 + k).

TF-IDF + LR reaches 0.763 on test; 0.620 with 750 labels

System	Test macro-F1	OFF P	OFF R	Spread
Always “not offensive”	0.444	—	0.000	deterministic
TF-IDF + LR, 750	0.620 ± 0.012	—	—	sd over 10 samples
TF-IDF + LR, 30,000	0.763	0.589	0.673	95% CI [0.745, 0.779]

The frozen test set was scored once for this deterministic baseline. All method results stay on dev until the final freeze.



Solid: TF-IDF dev. Dashed: TF-IDF test. Diamonds: fine-tuned BERTurk dev.

What this means. At 750 labels the cheap baseline is 0.620 on test and 0.613 on dev: the floor BERTurk and the LLMs must beat. With 30,000 labels TF-IDF reaches 0.763. BERTurk reaches 0.767 at 750 labels and 0.832 at 30,000 on dev.

275 dev errors; a 60-error audit shows where a word model fails

False alarms (166) — category	n
No insult / spurious cue	9
Political or religious criticism	10
Sports criticism or banter	5
Personal criticism	3
Sarcasm or irony	1
Label noise / ambiguous	2

Missed offensive (109) — category	n
Implicit insult, no profanity	8
Masked or abbreviated profanity	5
Veiled threat or violence	3
Political or religious insult	4
Institutional or sports criticism	3
Label noise / not offensive	7

Counts are in the 60-error audit (30 per direction, seed 42); each table sums to 30.

FALSE ALARM · F3 · 13105

“Aziz Yıldırım Fenerbahçe’yi padişahlık zannediyor herhalde…”

Club criticism without a slur.

MISS · M1 · 43325

“…bundan daha parazit canlı türü yok”

Implicit insult with no swear word.

MISS · M2 · 15069

“…bakanlık seçimi kaldı amkk”

Masked profanity, unseen by the word/char model.

Lesson: false alarms cluster on conflict and club vocabulary and character n-grams; misses cluster on implicit insults and masked profanity. About one in seven audited errors is label noise, so the ceiling is below 1.0.

Both arms run end-to-end; first dev numbers

Controlled change: baseline trains weights on the 750 labels; the LLM uses the identical 750 labels to pick 8 examples and a threshold. Nothing else differs except how labels are used.

LLM arm

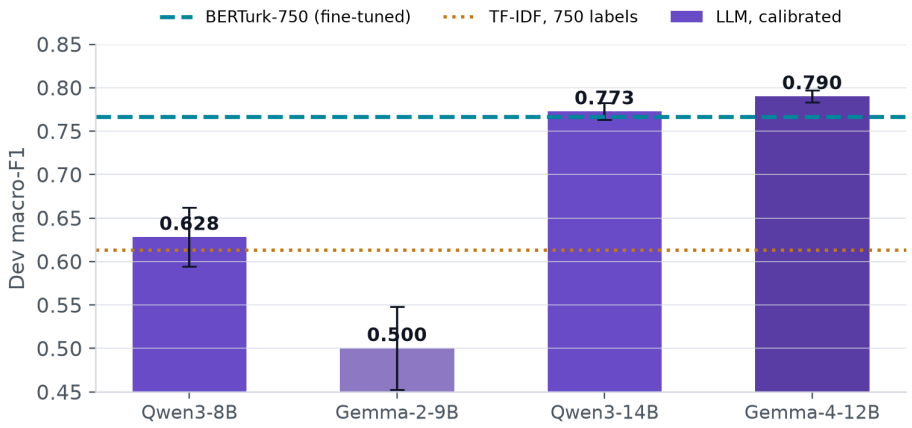
750 labels → 8 prompt examples (4 OFF / 4 NOT) → 742 labels set the threshold → score dev as log P(A) − log P(B).

Fine-tuning arm

BERTurk and XLM-R-base, same subsets: 128 tokens, batch 16, lr 2e-5, 4 epochs or 400 updates, no early stopping. 46 runs.

System (dev)	Macro-F1	Runs	Threshold
Qwen3-8B, zero-shot	0.162	1	0
Qwen3-8B, 8 + 742	0.628 ± 0.034	3	3.13 / 2.75 / 2.50
Gemma-2-9B, zero-shot	0.412	1	0
Gemma-2-9B, 8 + 742	0.500 ± 0.048	3	0.63 / 0.25 / 0.88
BERTurk, 750	0.767 ± 0.014	10	—
BERTurk, 30,000	0.832 ± 0.003	3	—

BERTurk beats both proposal LLMs; Gemma-4-12B leads on dev



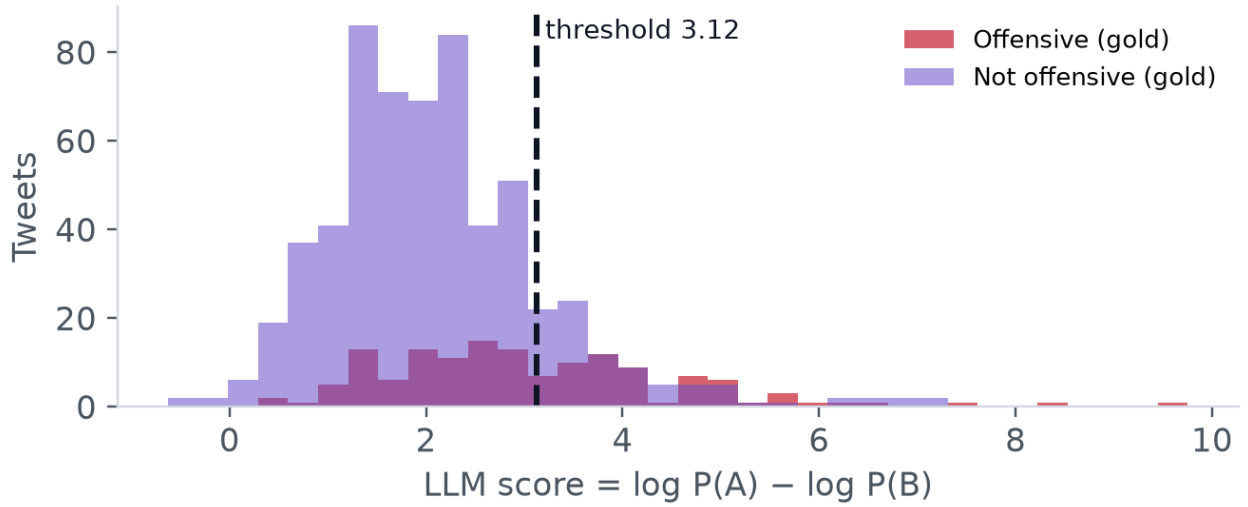
System	Dev macro-F1	Runs
TF-IDF + LR, 750	0.613 ± 0.015	10
Qwen3-8B	0.628 ± 0.034	3
Gemma-2-9B	0.500 ± 0.048	3
Qwen3-14B	0.773 ± 0.010	3
Gemma-4-12B	0.790 ± 0.007	3
BERTurk, 750	0.767 ± 0.014	10
BERTurk, 30,000	0.832 ± 0.003	3

BERTurk-750 is +0.154 over TF-IDF and +0.139 / +0.267 over Qwen3-8B / Gemma-2-9B. Gemma-4-12B is +0.023 over BERTurk-750. Test comparison pending.

Qwen's scores are shifted; Gemma's threshold is already near optimal

Zero-shot diagnostic	F1 @ 0	F1 @ best	AUC	mean OFF	mean NOT
Qwen3-14B	0.171	0.677	0.756	2.79	2.10
Gemma-4-12B	0.763	0.765	0.864	1.03	-4.13

F1 @ 0 uses threshold 0; F1 @ best is the best threshold. Mean OFF/NOT are mean log P(A) − log P(B). Chart: Qwen3-8B, one run.



What explains the gap

- **Not the tokenizer.** Gemma is more compact on these tweets: 32.3 vs 37.9 tokens/tweet.
- **Qwen is shifted, not blind.** At threshold 0 it calls 99% of tweets offensive, but the ranking is usable: best-threshold F1 is 0.677.
- **Gemma is already centred.** F1 at 0 is 0.763 and best is 0.765, so the 742 labels add only +0.027.
- **The 8 examples still help Qwen:** 0.677 → 0.773 after calibration.

This is threshold geometry, not proof of contamination. It does not isolate pretraining language from alignment; the exact-completion check is still planned.

One dev pass on an L4: 3–5 tweets/s with examples, 10–18 without

System	Params	Dev scoring	Throughput	Peak VRAM	Notes
TF-IDF, 750	—	0.6 s	~3,100/s	CPU	fit 1.0 s
TF-IDF, 30,000	—	0.6 s	~3,100/s	CPU	fit 14.6 s
Qwen3-8B, few-shot	8.2B	355 s	5.0/s	—	3 runs
Gemma-2-9B, few-shot	9.2B	430 s	4.1/s	—	3 runs
Qwen3-14B, few-shot	14.8B	644 s	2.7/s	—	3 runs
Gemma-4-12B, few-shot	12.0B	579 s	3.0/s	—	3 runs
Qwen3-14B, zero-shot	14.8B	180 s	9.8/s	10.7 GB	load 157 s · 1,290 tok/s

Scoring the 1,756 dev tweets; model download/load excluded unless stated. Few-shot is 3–4× slower because each prompt adds 8 examples (~600 tokens).

BERTurk: 35 s training at 750 labels, ~29 min at 30,000. Gemma-4-26B-A4B exceeded 22 GB and could not load on L4. Reported runs: 8.8 Colab units (~2.3 L4-hours).

BERTurk and Ablation A are done; remaining steps have a date

Date	Experiment	Status
11 Nov	BERTurk 750 (10) + 30,000 (3) on dev	done
11 Nov	Ablation A: BERTurk vs XLM-R, 750	done
12 Nov	Mid-project talk; freeze this deck	talk
21–24 Nov	Ablation B: Gemma with English instructions	planned
25–27 Nov	Test-contamination check, 200 test tweets	planned
28–30 Nov	Freeze methods; open test; final runs	planned
1–5 Dec	Paired bootstrap, error analysis, paper	planned
8 Dec	Final report	planned

Biggest risk

A 26B+ model may not fit the L4: Gemma-4-26B-A4B OOM'd at 22 GB, so we used the 12B dense model. **Fallback:** Qwen3-14B alone.

Result risk

The lines may never cross. The 750-label question still has a clear answer; the learning curve is descriptive.

Response to proposal feedback: the rubric asks for test-set baseline numbers, so we scored the deterministic TF-IDF baseline once on test and kept methods on dev.

What we know, what we do not, and what comes next

BASELINE · TEST

0.763

TF-IDF test macro-F1 at 30,000 labels; at 750 labels it falls to 0.620 ± 0.012. Deterministic; test 95% CI [0.745, 0.779].

BEST LLM · DEV

0.790

Gemma-4-12B dev macro-F1 after calibration; zero-shot 0.763. Qwen3-14B reaches 0.773. Threshold from 742 labels; 3 runs.

FINE-TUNE · DEV

0.767

BERTurk-750 dev (10 runs); 30,000 labels reaches 0.832. It beats Qwen3-8B and Gemma-2-9B, but Gemma-4-12B leads on dev. Gap to Gemma-4-12B: −0.023.

What we do not have yet: any test-set method result. BERTurk-750 is 0.767 on dev; the final answer uses test. Pre-registered before results: the frozen test set, macro-F1, the 0.02 minimum gap, and the paired bootstrap.

Questions welcome — any member can answer any part.

Fields 1–3: title and abstract, research question, contributions

1 · Title and abstract

Working title: Fine-tuning BERTurk vs threshold-calibrated open LLMs for Turkish offensive language with 750 labels.

Small Turkish platforms can label only a few hundred tweets. We ask whether fine-tuning BERTurk on 750 labelled OffensEval-TR tweets beats an open LLM that uses the same 750 labels as 8 prompt examples and 742 calibration tweets for its decision threshold. The TF-IDF + logistic-regression baseline scores 0.763 macro-F1 on the frozen 3,528-tweet test set, but 0.620 with 750 labels. The LLM arm scores each tweet as log P(A) − log P(B); the 742 labels set its threshold. On dev, calibration moves Qwen3-8B from 0.162 to 0.628 and Gemma-2-9B from 0.412 to 0.500; Qwen3-14B reaches 0.773 (zero-shot 0.171) and Gemma-4-12B 0.790 ± 0.007 (zero-shot 0.763). Fine-tuned BERTurk reaches 0.767 ± 0.014 on dev with 750 labels. The frozen-test comparison is pending; the final result uses the paired bootstrap.

2 · Research question

Unchanged. With 750 labels, does fine-tuning BERTurk beat prompting Gemma-2-9B-it or Qwen3-8B using the same 750 tweets (8 examples, 742 threshold labels) on macro-F1 over the official 3,528-tweet test set?

"No" result: BERTurk's mean over 10 runs is not ≥0.02 above the LLM's mean over 3 runs, or the paired bootstrap says chance ($p \geq 0.05$). **Change:** none.

3 · Contributions

C1. At 750 labels, we test whether BERTurk's mean test macro-F1 is ≥0.02 above each LLM's and whether a paired bootstrap rejects chance. Test pending; on dev BERTurk 0.767 vs 0.628 / 0.500.

C2. The 742-label threshold improves each LLM's dev macro-F1 over zero-shot; we report per-model delta and spread.

C3. The learning curve tests where fine-tuning overtakes TF-IDF between 250 and 6,000 labels; if the lines never cross, the 750-label answer stands.

Fields 4–5: related work and data

4 · Related work

1. **Zampieri et al. 2020**, SemEval-2020 Task 12. Top Turkish systems fine-tune on full data; no label-budget comparison.
2. **Çöltekin 2020**, LREC. The dataset; no fine-tuning-vs-prompting comparison.
3. **Zampieri et al. 2023**, NLE. Zero-shot LLMs; Turkish only Flan-T5-large 0.451; no examples.
4. **Dehghan & Yanıkoğlu 2024**, CASE. ChatGPT with 7 examples vs BERTurk, one dataset and one size.
5. **Pecher, Srba & Bielikova 2025**, EMNLP. English break-even near 100 labels; we test noisy Turkish two-class data.
6. **Schmitt & Schweter 2025**, SindBERT. Turkish encoders, full data only; no LLMs or small budgets.
7. **Şeker et al. 2025**, HATECAT-TR. Turkish span hate; BERTurk 0.59 vs LLaMA-3.3-70B 0.53; we classify whole tweets.
8. **Karakaş & Şimşek 2026**. Zero-shot LLMs vs BERTurk at one size; we add examples, calibration and a curve.

5 · Data

OffensEval-TR 2020, Çöltekin, LREC 2020; annotations CC BY 2.0; Turkish; @USER/URL masked. No extra cleaning.

Split	Tweets	OFF	OFF %
Official train	31,756	6,131	19.3
Our train	30,000	5,792	19.3
Dev	1,756	339	19.3
Test (frozen)	3,528	716	20.3

9 duplicate train texts; 1 test/train overlap. Example (OFF, id 33518): “Kırmızı ağır karar Allah belanı versin”.

Fields 6–9: method, protocol, results so far, error analysis

6 · Method and baseline

TF-IDF word 1–2 + char_wb 2–5, min_df=2, sublinear, Turkish lowercasing; LR C=1.0, balanced. Test 0.763 [0.745, 0.779].

Fine-tuning: BERTurk/XLM-R, 128 tokens, batch 16, 2e-5, 4 epochs or 400 updates, class-weighted, no early stopping. LLM: Qwen/Gemma 4-bit, 8 examples, 742-label threshold. One change: labels train weights vs choose examples and threshold.

Compute: 2.4 GPU-h fine-tuning (46 runs) + 2.5 L4-h for 16 LLM passes.

7 · Evaluation protocol

Macro-F1; frozen 3,528-tweet test; 10 BERTurk runs at 250/750/2,000 and 3 at 6,000/30,000; 3 runs per LLM at 750. Paired bootstrap, 10,000 resamples; real gap ≥0.02 and wins on ≥95%. Fixed before method results; dev for code checks only.

8 · Results so far (mean ± sd) — BERTurk-30k = 0.832 ± 0.003

System	Dev	Test	System	Dev	Test
Always NOT	0.447	0.444	Gemma-2-9B zero	0.412	—
TF-IDF, 750	0.613 ± 0.015	0.620 ± 0.012	Gemma-2-9B cal	0.500 ± 0.048	—
TF-IDF, 30,000	0.763	0.763	Gemma-4-12B zero	0.763	—
Qwen3-8B zero	0.162	—	Gemma-4-12B cal	0.790 ± 0.007	—
Qwen3-8B cal	0.628 ± 0.034	—	BERTurk, 750	0.767 ± 0.014	—
Qwen3-14B zero	0.171	—	BERTurk, 30,000	0.832 ± 0.003	—
Qwen3-14B cal	0.773 ± 0.009	—	XLM-R, 750	0.670 ± 0.025	—

9 · Error analysis — 60 audited of 275

False alarms: no insult 9; political/religious 10; sports 5; personal 3; sarcasm 1; label noise 2. Misses: implicit insult 8; masked profanity 5; veiled threat 3; political/religious 4; institutional/sports 3; label noise 7. One example per category is in results/errors/audit.json.

Fields 10–12: research lead, plan and risk, reproducibility

10 · Research lead section — Vesile Han (CSE 654), fine-tuning; Gürkan Göztepeli (CSE 654), prompting

Vesile Han. Pecher et al., EMNLP 2025: English break-even near 100 labels, more for two-class tasks and variance; we test Turkish noisy two-class. Schmitt & Schweter, SindBERT 2025: Turkish encoders, full data only; no LLMs. Ablation A (done): BERTurk 0.767 vs XLM-R 0.670, +0.097 [0.080, 0.116], 100% wins. KUISAIL published BERTurk 0.816 on test; our 30k run should reproduce.

Gürkan Göztepeli. Şeker et al., HATECAT-TR 2025: BERTurk 0.59 vs LLaMA-3.3-70B 0.53 on Turkish hate spans; we classify whole tweets and score probabilities. Karakaş & Şimşek 2026: zero-shot LLMs vs BERTurk at one size; we add examples, calibration, curve. Ablation B: Gemma English instructions, 3 runs.

11 · Plan and risk

Done: BERTurk curve (36 runs), Ablation A (10). 21–24 Nov: Ablation B. 25–27 Nov: contamination check. 28–30 Nov: freeze, open test, final runs. 1–5 Dec: bootstrap, error analysis, paper. 8 Dec: report. Owners to be confirmed.

Biggest risk: 26B+ may not fit L4; Gemma-4-26B OOM'd at 22 GB, so 12B used. Fallback: Qwen3-14B alone. **Result risk:** lines may never cross; 750-label answer stands.

12 · Reproducibility

Repository: github.com/paysis/offenseval-tr-midproject. One command: `bash scripts/reproduce.sh gpu`. Seeds: split 42, subsets 42+k, bootstrap 42. Versions: Python 3.13.16, scikit-learn 1.9.1, transformers 5.22.1, torch 2.14.1, bitsandbytes 4-bit NF4.

Roles: to be confirmed before the final report. **AI tools:** a coding assistant was used for code, analysis and drafting; every number is read from saved result files and checked by the team.